

Section 4.4: Network Devices

In order to build a network, you need to use different types of **hardware**.

In this section we discuss these types of hardware and describe exactly what they are used for.

Key Concepts of this section:

- # Understand that there are **seven** networking devices that can be used to construct a computer network.
- # Know the **features** of each network device.
- # Be able to **describe** the **role** that each networking device plays in the construction of a network.

Types of networking devices

There are **seven** main devices that are used to build a network. These are:

- [Router](#)
- [Network Interface Cards \(NIC\)](#)
- [Network Cable](#)
- [Hubs](#)
- [Switches](#)
- [Bridges](#)
- [Modems](#).

We will discuss each type of device and look at what they are used for below:

1 Router

What are routers used for?

Routers enable **data** to be **sent** (routed) between **different** types of **networks**.

For example:

A router could be used to connect a **LAN** (local area network) to a **WAN** (wide area network).

Routers are most commonly used to **connect computers** and **devices** to the **internet** (WAN).

Computers can connect to a router either through cables or wirelessly.

What exactly do routers do?

At this point, you can see that routers can **connect** different types of **networks** together and **send data** between them.

They can do this because they are **intelligent** devices and can perform the following functions:

- They can **read data** and decide **where** to send it
- They can decide on the **fastest route** in which to send the data
- They can make the **format** of the data **suitable** for the network where it is being sent.

This is where the name '**router**' comes from. They can **direct** data between networks using the **best/fastest route** possible.

IP Addresses

All computers and devices connected to the internet are assigned a **unique number** called an **Internet Protocol Address** (IP address).

A computer or device's IP address determines it's **exact location**.

The IP address of the device would depend on **where in the world** it connected to the internet from.

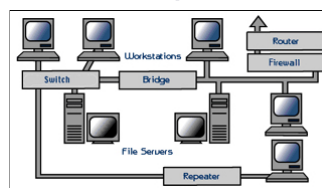
The list below shows some examples of different IP addresses in various countries:



Key Words:

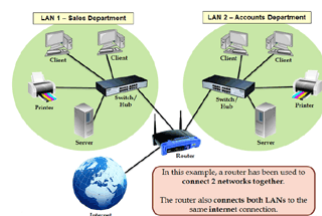
Router, Network cards, Cable, Hub, Bridge, Switch, Modem.

Examples:

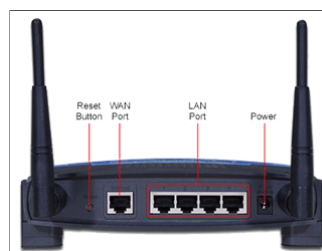


Networks consist of various hardware.

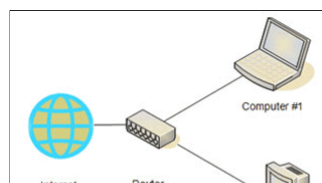
Examples:



Routers are used to connect networks together or to connect networks to the internet.
(click to zoom)



Routers allow computers on a LAN to share the same internet connection.



- **Bermuda** - 64.147.80.0
- **United Kingdom** - 80.247.16.0
- **United States** - 168.99.0.0

- # Computers on the **same network** would share the **same first few numbers** of the IP address.

This is used to **locate** the **network**.

- # Different **devices** on the same network are uniquely identified by the **last few numbers** of the IP address.

On my school's network, IP address for individual devices could be assigned like this:

- **Desktop PC** - 64.128.80.101
- **Printer** - 64.128.80.147

This is used to **locate** individual **devices** on the network.

How does a router store IP addresses?

- # Routers store IP addresses in something called a **routing table**.
- # The routing table **lists** all of the **different routes** to other networks.
- # The router will use the routing table to determine the **best route** to use when sending data to another network.

How does a router send data between computers?

- # Routers send '**packets**' of data between computers on different networks.
- # Each data packet contains the **IP address** of the computer/network that the data is being sent to.
- #

The router will use the IP address to work out the **best route** in which to send the data to it's destination.

Remember:

The router will use the **first few numbers** of the IP address to determine the **location** of the **network** and the **last numbers** to determine which **device** on the network has requested the data.

- # The data will be received by **routers** on other networks which will read the IP address and **re-route** the data until it ends up at the **exact device** it was intended for.

2 Network Interface Card (NIC)

What are network interface cards used for?

- # Network Interface Cards are used to **connect** individual **computers/devices** to a **network**.

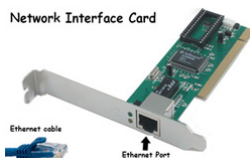
Modern computers usually come with network interface cards already **built-in**.

- # Network interface cards have **ports** which allow **network cables** to be **plugged in** and connect the computer to the network.

Note:

There are **two** types of network interface card:

- Wired network interface card (Where cables are used to connect computers)
- Wireless network interface card (Where computers are connected using Wi-Fi)



A wired network interface card connects computers to a network using cables.



A wireless network interface card connects computers to a network using Wi-Fi signals.



Computer #2

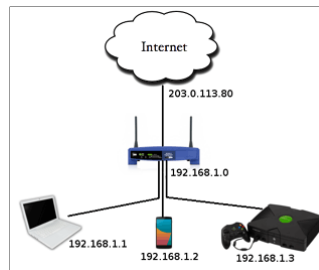
Multiple devices connecting to the internet through the same internet connection.

```
[root@fedora10 ~]# netstat -nr
Kernel IP routing table
```

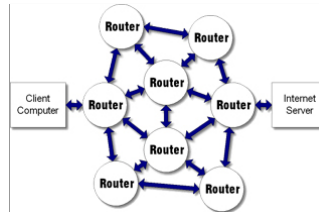
Destination	Gateway	Genmask	Flags
60.49.199.72	0.0.0.0	255.255.255.240	U
172.16.163.0	172.16.160.1	255.255.255.0	UG
172.16.162.0	172.16.160.1	255.255.255.0	UG
172.16.161.0	172.16.160.1	255.255.255.0	UG
172.16.160.0	0.0.0.0	255.255.255.0	U
172.16.167.0	172.16.160.1	255.255.255.0	UG
172.16.166.0	172.16.160.1	255.255.255.0	UG
172.16.165.0	172.16.160.1	255.255.255.0	UG
172.16.164.0	172.16.160.1	255.255.255.0	UG
172.16.170.0	172.16.160.1	255.255.255.0	UG
172.16.169.0	172.16.160.1	255.255.255.0	UG
172.16.168.0	172.16.160.1	255.255.255.0	UG
169.254.0.0	0.0.0.0	255.255.0.0	U
169.254.0.0	0.0.0.0	255.255.0.0	U
0.0.0.0	60.49.199.72	0.0.0.0	UG

```
[root@fedora10 ~]#
```

Routers store IP addresses in a routing table which they use to work out the best route in which to send data to it's destination network.

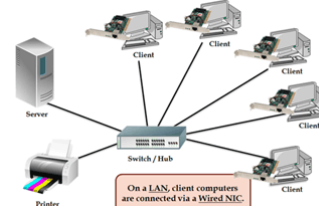


Routers use IP addresses to work out where to send packets of data.

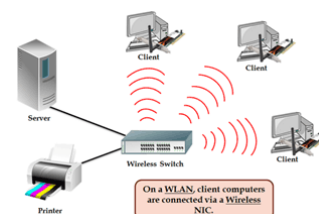


Packets of data sent over the internet can pass through many other network's routers until it reaches it's destination.

Examples:



A wired NIC uses cables to connect computers/devices to a network. (click to zoom)



A wireless NIC uses Wi-Fi to connect computers/devices to a network. (click to zoom)

3 Network cable

What are network cables used for?

- # Network cables are **plugged** into a computer's **wired network interface card** and connects it to the network.

Data is sent around the network via the network cable.

- # Cable holds **advantages** over wireless connections for two reasons:
 - Cables can transfer data **faster** than wireless
 - Data transferred over cables is **more secure** than over wireless (**Hackers can't intercept data easily**).

Note:

Network cables are made up of different wires:

- Some wires are used to **send data to** the computer
- Some wires are used to **receive data from** the computer.



A network cable.



Network cables have connectors on each end that plug into network interface cards.



The wires inside a network cable perform different functions.

Examples:



Network cables plug into the ports on network interface cards.



Network cable is used to connect computers and devices to a network.

4 Hubs

What are hubs used for?

- # Hubs allow computers and devices to plug into their ports in order to **connect** to each other and **share files, data** and **resources**.
- # Hubs are '**non-intelligent**' devices and they **don't manage** any of the data that flows through them.

When data gets sent to the hub, there is **no attempt** to **locate** the computer/device that the data is **meant for**.

The hub simply sends the data onto **every** computer/device on the network.

This means that every device on the network will **receive** the **same data** whether they requested it or not.

- # This lack of data management makes networks that are connected by hubs **very slow** because there is a lot of **unnecessary data** flowing around.

Note:

Hubs are **old technology** and have been replaced by switches which manage data more effectively and operate much faster (**more on switches later**).

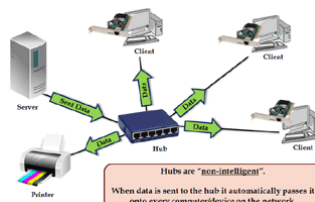
See the image to the right for a picture example of how hubs work



Examples:



Hubs allow devices to connect to each other by plugging network cables into their ports.



Hubs are 'non-intelligent' devices and do not manage data flowing around the network. (click to zoom)

5 Switch

What are switches used for?

- # Switches are **similar to hubs** in that they connect computers/devices to form a LAN.
- However, switches are '**intelligent**' devices and transmit data around the network **more efficiently**.

How do switches manage network data?

- # How switches manage data is summarised below:
 - Each network device has a **Media Access Control (MAC) address** which uniquely identifies it.
 - Data sent to the switch contains the MAC address of the **sending device** and the MAC address of the **receiving device**.
 - The switch will **check** these addresses and only send the data to the **relevant device** rather than to all devices.

See the image to the right for a picture example of how switches work

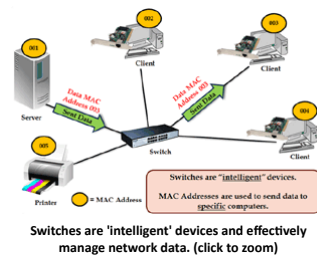


Examples:



A network switch works in a similar way to a hub in that it allows network devices to connect to each other.

Because switches send data directly to the correct device, the amount of unnecessary data traveling around the network is **reduced** and the whole network **works faster**.



6 Bridge

What are bridges used for?

- # Bridges are used to **connect** (bridge) LAN's together.
- # Bridges can **connect** different types of **networks** so that they act as **one single LAN** and thus can communicate with each other.

For example:

You have **two networks**. One network connects the computers/devices with **cables** and the other connects the computers/devices using **wireless**.

The cabled network **cannot communicate** with the wireless network and vice versa.

By using a bridge, all of the computers/devices can communicate with each other as **one single LAN**.

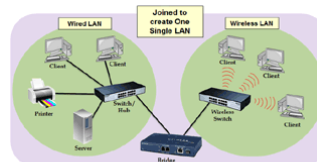
See the image to the right for a picture example of how bridges work



Examples:



Bridges are used to connect (bridge) multiple LAN's together.



A bridge can allow two different LAN's to communicate with each other. (click to zoom)

7 Modems

What are modems used for?

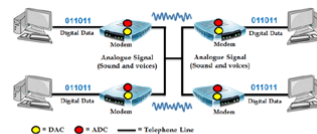
- # Modem stands for '**Modulator Demodulator**'.
- # Modems were used to connect computers to the internet before we had the use of broadband connections.

NOTE:

Some modern modems have routers **built-in**. This allows **multiple devices** to connect to the **same** internet connection.

If your modem does not have a built-in router, you will only be able to connect **ONE** device to the internet at a time. You can, alternatively, buy a **separate router** and connect it to your modem.

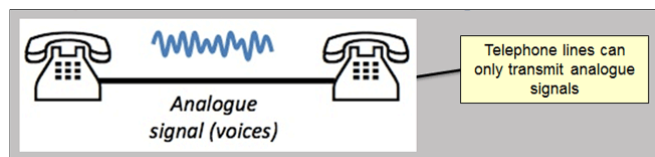
Examples:



This diagram shows how modems are used to convert data between analogue and digital in order to send down a telephone line. (click to zoom)

How do modems work?

- # Most internet connections are made over **telephone lines**. Telephone lines are designed to carry **sound and voices**, which are **analogue** signals.



- # The problem is that computer data is **digital** and it is **not possible** to send digital data over an analogue telephone line.

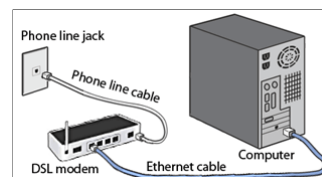


- # This is where the modem comes in.

The modem can **convert** the **digital** computer data into an **analogue** signal/noise (**modulate**) so it can be sent over the analogue telephone line.



An old modem which allowed people to connect to the internet.



(modulate) so it can be sent over the analogue telephone line.

- # Modems can also reverse this process and convert the **analogue** signals from the telephone line into **digital** data (**demodulate**) so the computer can use it.
- # Modems contain both a **Digital to Analogue Converter** (DAC) and a **Analogue to Digital Converter** (ADC).

Note:

DAC
(modulator)



Converts the **digital data** from the computer into an **analogue signal** which can then be sent down the telephone line.

(Literally converts digital data into noises which is what the telephone line is designed to carry - [demo](#))

ADC
(Demodulator)



Converts the **analogue signal** in the telephone line into **digital data** so that the computer can process and use the data.



Modems convert digital signals into analogue (sound) so they can be sent down a telephone line to their destination.



Some modems have routers built-in and some don't. If you want to connect multiple devices to the same internet connection you will need both.

Activity!

[Network devices - Research Sheet](#)

Click the above task and answer the questions about the different types of hardware that are needed to build a network.



[Previous - Types of Networks](#)

[Next - Internet and Intranet](#)



Please add your questions/comments below:

0 Comments [ictlounge](#)

Login

Recommend 6 [Share](#)

Sort by Best



Start the discussion...

Be the first to comment.

Subscribe [Add Disqus to your site](#) [Add Disqus Add](#) [Privacy](#)

Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

[Section 4: Networks and the Effects of using them](#)

[Section 7: The Systems Life Cycle](#)

Section 10: Communication

[Section 2: Input and Output Devices](#)

[Section 5: The Effects of using ICT](#)

[Section 8: Safety and Security](#)

[Section 3: Storage Devices and Media](#)

[Section 6: ICT Applications](#)

[Section 9: Audience](#)

Links to Practical Units:

Section 11: File Management

Section 14: Styles

Section 17: Document Production

Section 20: Data Analysis

Section 12: Images

Section 15: Proofing

Section 18: Data Manipulation

Section 21: Website Authoring

Section 13: layout

Section 16: Graphs and Charts

Section 19: Presentations

[Back to top](#)